Prepare Fold-Local Spatial Predictors

Description

Computes Moran eigenvector predictors from training locations or samples only and projects them to held-out samples with the production IDW path.

Usage

```
prepare_fold_spatial_predictors(
  data_coords_projected = NULL,
  data_sample_ids = NULL,
  train_ids = NULL,
  test_ids = NULL,
  spatial_mode = NULL,
  n_mev = NULL,
  spatial_mev_config = NULL,
  compute_spatial_function = compute_spatial_mev,
  compute_spatiotemporal_function = compute_spatiotemporal_mev,
  interpolate_spatial_function = interpolate_mev_to_grid,
  interpolate_spatiotemporal_function = interpolate_st_mev_to_grid,
  compute_spatial_basis_function = compute_spatial_mev_basis,
  project_spatial_basis_function = project_spatial_mev_basis
)
```

Arguments

`data_coords_projected`	Data frame with site row names and 'coord_x_km' and 'coord_y_km' columns.
`data_sample_ids`	Data frame containing 'dataset_name' and 'age' for all candidate samples.
`train_ids`, `test_ids`	Disjoint sample identifiers in '"dataset_name__age"' format. Locations represented by the two partitions must

	also be disjoint.
`spatial_mode`	Character scalar. Supported values are '"spatial"' and '"spatiotemporal"'.
`n_mev`	Positive integer giving the requested number of Moran eigenvectors.
`spatial_mev_config`	Optional shared 2-D MEM strategy configuration. When 'NULL', the legacy compute and interpolation functions are retained for compatibility tests.
`compute_spatial_function`, `compute_spatiotemporal_function`	Functions used to compute 2-D and spatiotemporal Moran eigenvectors.
`interpolate_spatial_function`, `interpolate_spatiotemporal_function`	Functions used to project training Moran eigenvectors to held-out samples.
`compute_spatial_basis_function`, `project_spatial_basis_function`	Functions used to construct and project an explicit reusable 2-D basis.

Value

Named list with unscaled training and held-out spatial predictor data frames plus fold-level spatial diagnostics, including requested, available, and unavailable Moran eigenvector counts.

Examples

```
## Not run:

list_spatial_fold <-

  prepare_fold_spatial_predictors(

    data_coords_projected = data_coords,

    data_sample_ids = data_samples,

    train_ids = vec_train_ids,

    test_ids = vec_test_ids,

    spatial_mode = "spatial",

    n_mev = 3L

  )
```

```
## End(Not run)
```